

A Non-Split Regge–Wheeler Self-Extension and a Defective Schwarzschild Resonance in Pure Weyl Gravity

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Abstract

We determine the differential-module and resonance structure of axial Schwarzschild perturbations in four-dimensional pure Weyl gravity. The six-state Bach system has an exact filtration with successive quotients given by two spin-two Regge–Wheeler modules and one spin-one Maxwell module. The repeated spin-two block is a non-split self-extension and is the first jet of a four-state spin-two/spin-one family.

After Liouville reduction, the extension is represented by a scalar projective cocycle modulo the symmetric-square operator

$$\mathcal{K}_U = D^3 + 4UD + 2D(U).$$

For axial $\ell = 2$ we prove the exact normal form

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} \left[1 - \frac{2}{r} \right].$$

Thus the Bach extension is locally cohomologous to an elementary mass-shaped direction, up to parameter normalization and endpoint frames.

At a simple spin-two quasinormal frequency the complete connection admits two possible local Smith types. A validated projective Evans calculation selects the defective type $(0, 0, 2)$ at one damped Schwarzschild mode. The geometric root vector is Einstein, whereas its generalized root vector has nonzero projection to the Ricci-carrier quotient. We further show that the compact-source, locally observed outgoing Green operator for the reduced axial system has a nonzero rank-one pole of order two at this frequency.

Promotion to a global physical resolvent and to a time-domain generalized ringdown term requires a separate analytic Fredholm realization, weighted endpoint domains, and bounded metric reconstruction.

1 Introduction and main results

Pure Weyl gravity has a fourth-order metric equation, but its Schwarzschild perturbations are not best understood as an unstructured fourth-order system. In the axial $\ell = 2$ sector the six-dimensional radial system has the filtered form

$$0 \subset M_2 \subset E_2 \subset M_{\text{Bach}}, \quad \text{gr } M_{\text{Bach}} \simeq M_2 \oplus M_2 \oplus M_1,$$

where M_2 is the spin-two Regge–Wheeler module and M_1 is the spin-one Maxwell module. The middle object E_2 is a self-extension

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0.$$

The decisive point is that this extension is non-split over the rational differential field. The two spin-two layers therefore share scalar characteristics without defining two rationally separable radial branches.

This paper establishes three central results.

1. The repeated spin-two block is a non-split Regge–Wheeler self-extension, realized exactly as a parameter jet.
2. Its projective extension class has the elementary normal form

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2} \left[1 - \frac{2}{r} \right].$$

3. At a certified damped Schwarzschild quasinormal frequency the connection is defective. The associated generalized root vector has a nonzero carrier quotient, and the reduced outgoing Green operator has a genuine rank-one double pole.

The third result is stronger than a statement about a finite connection matrix. It is a meromorphic differential-response theorem for compact radial sources and local radial observations. It is deliberately weaker than a causal spacetime resolvent theorem: no inverse-Laplace deformation, late-time expansion, or $te^{i\omega_n t}$ assertion is made here.

Logical status

Result	Required input	Status
RW/RW/Maxwell filtration and partial jet	Exact rational identities and reconstruction rank	Exact
Non-split spin-two extension	Exhaustive rational coboundary obstruction	Exact for real $\omega > 0$
Cocycle mass-direction normal form	Exact substitution in $\mathbb{C}(r, \omega)$	Exact for $\omega \neq 0$
Defective physical specialization	Winding-one contour, local selector, spin-one unit	Validated theorem
Non-Einstein generalized root	Defective Smith type and exact filtration	Exact consequence
Reduced outgoing Green double pole	Analytic endpoint maps and finite-interval reduction	Exact/validated theorem
Causal spacetime pole and generalized ringdown	Global weighted Fredholm and contour theory	Not claimed
Physical Einstein–Weyl mass identification	Finite-mass cocycle and endpoint normalization	Open

2 Differential-module conventions

We work over

$$K = \mathbb{C}(r, \omega), \quad D = f\partial_r, \quad f = 1 - \frac{2}{r},$$

with $\omega \neq 0$ unless a threshold limit is stated explicitly. All rational gauges and module morphisms are over the differential field (K, D) . We use $D(q)$, $D(U)$, and D^3q , rather than primes, to avoid confusing the tortoise derivative with ∂_r .

The axial spin-two and spin-one scalar operators are

$$L_2 = D^2 + U_2, \quad U_2 = \omega^2 - f \left(\frac{6}{r^2} - \frac{6}{r^3} \right),$$

$$L_1 = D^2 + U_1, \quad U_1 = \omega^2 - f \frac{6}{r^2}.$$

The latter is the $\ell = 2$ Maxwell Regge–Wheeler equation. A rank-two scalar module is represented by its companion system on $(y, Dy)^T$.

The source Einstein layer is the kernel of the linearized Ricci/Schouten carrier map. The target spin-two quotient is its physical Regge–Wheeler carrier. The third quotient is the Maxwell class of the target Einstein gauge vector. Consequently, “non-Einstein” below means nonzero projection to the gauge-invariant carrier quotient, not failure of a coordinate gauge condition.

3 Exact axial RW/RW/Maxwell filtration

In the certified factor gauge the six-state system is

$$\begin{aligned} DX &= AX + EY + CZ, \\ DY &= AY + DZ, \\ DZ &= A_1 Z, \end{aligned} \tag{1}$$

where $X, Y, Z \in K^2$, the companion matrices A and A_1 represent L_2 and L_1 , and the forcing blocks factor through a single scalar source:

$$E = USJ, \quad C = USN.$$

The rational reconstruction has full row rank on the declared domain.

Theorem 3.1 (Filtered axial Bach module). *For axial $\ell = 2$ Schwarzschild perturbations, the reduced six-state Bach module has a filtration whose successive simple factors are*

$$M_2, \quad M_2, \quad M_1.$$

The middle two-spin-two block is an extension E_2 of M_2 by M_2 . The quotient M_1 is the Maxwell module of the carrier gauge vector. The filtration is exact but is not asserted to be a direct sum.

The theorem is an exact rational statement. It is independent of endpoint normalization, scattering conventions, and the later QNM specialization.

4 Partial-jet realization

Introduce the four-state family

$$\mathcal{B}(\tau) = \begin{pmatrix} A + \tau E & D + \tau C \\ 0 & A_1 \end{pmatrix}.$$

Let its fundamental matrix be

$$\Phi_4(\tau) = \begin{pmatrix} P(\tau) & Q(\tau) \\ 0 & R \end{pmatrix}.$$

Theorem 4.1 (Exact partial jet). *The six-state generator in (1) is the first spin-two-row jet of $\mathcal{B}(\tau)$ at $\tau = 0$. Its fundamental matrix is*

$$\Phi_6 = \begin{pmatrix} P & \dot{P} & \dot{Q} \\ 0 & P & Q \\ 0 & 0 & R \end{pmatrix}_{\tau=0}, \quad \det \Phi_6 = (\det P)^2 \det R.$$

Consequently any factor-adapted endpoint connection has the triangular form

$$T(\omega) = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix}, \quad \det T = a^2 g.$$

Proof. Differentiate $D\Phi_4(\tau) = \mathcal{B}(\tau)\Phi_4(\tau)$ at $\tau = 0$. The equations for $P, \dot{P}, Q, \dot{Q}, R$ reproduce the three rows of (1) column by column. The determinant identity follows from block triangularity. The same identity applied to endpoint transition matrices gives the displayed connection form. $\square$

The parameter τ is intrinsic at this stage. Identifying it with a physical Fierz–Pauli mass requires an additional finite-mass reduction and compatible endpoint germs; Section 15 records the exact remaining gate.

5 Projective gauge and symmetric-square cohomology

Consider a scalar self-extension

$$Lx = (s_1 D + s_0)y, \quad Ly = 0, \quad L = D^2 + U. \quad (2)$$

Set

$$\mathcal{I} = s_0 - \frac{1}{2}D(s_1), \quad \mathcal{K}_U = D^3 + 4UD + 2D(U).$$

Lemma 5.1 (Explicit triangular gauge). *For $q \in K$, define*

$$Q(q) = qD - \frac{1}{2}D(q).$$

On every solution $Ly = 0$,

$$[L, Q(q)]y = -\frac{1}{2}\mathcal{K}_U(q)y.$$

Thus the change of lift $x \mapsto x + Q(q)y$ changes the scalar extension density by

$$\mathcal{I} \mapsto \mathcal{I} - \frac{1}{2}\mathcal{K}_U(q).$$

Proof. Expand $LQ(q) - Q(q)L$, use the Leibniz rule for D , and then set $D^2y = -Uy$. The coefficient of Dy cancels, while the coefficient of y is $-\frac{1}{2}[D^3q + 4UDq + 2D(U)q]$. $\square$

The extension is therefore represented by

$$[\mathcal{I}] \in K/\mathcal{K}_U K.$$

Because Lemma 5.1 contains the factor $1/2$, removal of a term $\mathcal{K}_U q$ uses the triangular gauge $Q(2q)$. This factor does not alter the cohomology quotient but matters in an explicit frame calculation.

The equation $\mathcal{K}_U q = 0$ is the symmetric-square equation of L . If y_1, y_2 span the scalar solution space, then its solution space is spanned by $y_1^2, y_1 y_2, y_2^2$. The same operator controls both the projective obstruction and trace-free horizontal endomorphisms.

6 Mass-direction normal form of the Bach cocycle

For the axial $\ell = 2$ Bach extension, rational reduction gives

$$\mathcal{I}_{\text{Bach}} = \frac{i(r-2)(2r\omega^2 + 3\omega^2 + 12)}{5r^4\omega}.$$

Theorem 6.1 (Mass-direction normal form). *Let*

$$q(r, \omega) = -\frac{i}{120\omega} \left(15r + 13 + \frac{12}{r} + \frac{9}{r^2} \right).$$

For $\omega \neq 0$,

$$\mathcal{I}_{\text{Bach}} = \mathcal{K}_{U_2}q + \frac{i\omega}{2}f.$$

Consequently

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2}[f] \quad \text{in} \quad K/\mathcal{K}_{U_2}K.$$

In particular, $[f] \neq 0$ wherever the Bach self-extension is non-split.

Proof. Apply $D = f\partial_r$ three times to q and substitute

$$U_2 = \omega^2 - f \left(\frac{6}{r^2} - \frac{6}{r^3} \right)$$

in $\mathcal{K}_{U_2}q = D^3q + 4U_2Dq + 2D(U_2)q$. Clearing the common denominator $120r^4\omega$ reduces the claim to a polynomial identity in r and ω . Exact coefficient comparison gives the displayed formula. The quotient identity follows immediately. If $[f] = 0$, then $[\mathcal{I}_{\text{Bach}}] = 0$, which would split the extension. $\square$

Remark 6.2 (Threshold scope). The displayed $1/\omega$ singularity of $\mathcal{I}_{\text{Bach}}$ is cohomologically removable: its intrinsic class has the representative $(i\omega/2)f$. The removing gauge $Q(2q)$, however, is itself $O(\omega^{-1})$. The theorem therefore does not construct a threshold-uniform splitting or justify commuting the limits $\omega \rightarrow 0$ and $\tau \rightarrow 0$.

Remark 6.3 (Mass-shaped is not yet physical mass). The representative $f = 1 - 2/r$ is the radial direction expected from a mass term after a helicity-two reduction. The theorem proves a local cohomology normal form, not equality with a particular Einstein–Weyl mass parameter. If a finite-mass reduction gives

$$\mathcal{I}_{\text{mass}} = c(\omega)f + \mathcal{K}_{U_2}p,$$

then the two extensions are locally equivalent precisely when $c(\omega) \neq 0$; their parameter normalization is $2c(\omega)/(i\omega)$.

7 Period formula for the connection shear

Let y_1, y_2 be a scalar basis and set

$$Y = \begin{pmatrix} y_1 & y_2 \\ Dy_1 & Dy_2 \end{pmatrix}, \quad W = y_1Dy_2 - y_2Dy_1.$$

Because the companion matrix has zero trace, $D(W) = 0$.

Theorem 7.1 (Symmetric-square period matrix). *For the family*

$$(D^2 + U + \tau V)y = 0,$$

the interaction-picture variation satisfies

$$Y^{-1} \partial_\tau A Y = \frac{V}{W} \begin{pmatrix} y_1 y_2 & y_2^2 \\ -y_1^2 & -y_1 y_2 \end{pmatrix}.$$

Consequently the spin-two connection shear is determined by the three periods

$$\int \frac{V}{W} y_1^2 dr_*, \quad \int \frac{V}{W} y_1 y_2 dr_*, \quad \int \frac{V}{W} y_2^2 dr_*.$$

Proof. In the basis $(y, Dy)^T$,

$$A(\tau) = \begin{pmatrix} 0 & 1 \\ -(U + \tau V) & 0 \end{pmatrix}, \quad \partial_\tau A = \begin{pmatrix} 0 & 0 \\ -V & 0 \end{pmatrix}.$$

Insert

$$Y^{-1} = \frac{1}{W} \begin{pmatrix} Dy_2 & -y_2 \\ -Dy_1 & y_1 \end{pmatrix}$$

and multiply. Integration of the interaction-picture equation gives the three entries. $\square$

A coboundary $\mathcal{K}_{Uq}$ changes these periods by endpoint terms. This is the connection-level form of Lemma 5.1: rational triangular gauges alter Jost frames but not the intrinsic divisor or Smith selector once the endpoint units are tracked.

8 Nonsplitting, endomorphisms, and Fitting ideals

Theorem 8.1 (Non-split physical-axis extension). *For every real $\omega > 0$, the axial $\ell = 2$ spin-two block*

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0$$

is non-split over $\mathbb{C}(r)[D]$.

Certificate-supported exact proof. The rational splitting equation reduces, after exhaustive pole and degree bounds, to a finite coefficient system. Fixed minors of its coefficient and augmented matrices are

$$\det M_{[0,1,2]} = 3456 \omega^{10}, \quad \det[M \mid \text{rhs}]_{[0,1,2,5]} = -645120 i \omega^9.$$

They are simultaneously nonzero for every real $\omega > 0$, so the inhomogeneous coboundary equation is inconsistent. The fixed-minor argument also covers the former frame event $\omega^2 = 3$. $\square$

The scalar Regge–Wheeler module is simple on the positive real axis and has scalar rational endomorphism ring. The only nonzero complex reducibility points allowed by the terminal Riccati ansatz with matching horizon and infinity signs are the algebraically special frequencies

$$\omega = \pm i \frac{(\ell - 1)\ell(\ell + 1)(\ell + 2)}{12}.$$

For $\ell = 2$, these are $\omega = \pm 2i$, away from the physical axis.

Theorem 8.2 (Local commutant). *For real $\omega > 0$,*

$$\text{End}_{\mathbb{C}(r)[D]}(E_2) \simeq \mathbb{C}[\varepsilon]/(\varepsilon^2).$$

Every endomorphism is $aI + bN$, where $N^2 = 0$. Hence the only idempotents are $0, I$, the only involutions are $\pm I$, and every diagonalizable local endomorphism is scalar.

Proof. The submodule M_2 is the unique simple submodule: a second simple submodule mapping nontrivially to the quotient would split the extension. An endomorphism therefore induces scalars a, d on the submodule and quotient. Naturality of the nonzero extension class gives $a = d$. Subtracting aI leaves a map that vanishes on the submodule and quotient, so it factors through the quotient back into the submodule and is a scalar multiple of the canonical nilpotent N . The corollaries follow from $(aI + bN)^k = a^k I + k a^{k-1} b N$. $\square$

For the triangular connection

$$T = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix},$$

the Fitting ideals are

$$I_1 = (a, b, c, d, g), \quad I_2 = (a^2, ad, bd - ac, ag, bg), \quad I_3 = (a^2 g).$$

They give a normalization-independent local classification of all specializations.

9 Analytic Jost connections and Smith dichotomy

The moving Frobenius/Jost phases define analytic endpoint-selected planes on the certified spectral domains. Transport to a common ordinary radius and change to a factor-adapted frame. At a simple spin-two QNM ω_n , assume the spin-one coefficient $g(\omega_n) \neq 0$. Analytic row and column operations remove the spin-one row and column, leaving

$$T_2(\omega) = \begin{pmatrix} a(\omega) & b(\omega) \\ 0 & a(\omega) \end{pmatrix}, \quad a(\omega_n) = 0, \quad a'(\omega_n) \neq 0.$$

Proposition 9.1 (Local Smith dichotomy). *At ω_n ,*

$$\text{Smith}(T) = \begin{cases} (0, 0, 2), & b(\omega_n) \neq 0, \\ (0, 1, 1), & b(\omega_n) = 0. \end{cases}$$

The first case has geometric multiplicity one and one length-two root chain; the second has two independent length-one roots.

Proof. Let $z = \omega - \omega_n$. Since $a = a_1 z + O(z^2)$ with $a_1 \neq 0$, the determinant valuation of the two-spin-two block is two. If $b(0) \neq 0$, its first Fitting ideal is the unit ideal, so its exponents are $(0, 2)$. If $b(0) = 0$, all entries vanish to first order and the exponents are $(1, 1)$. Appending the spin-one unit adds a zero exponent. $\square$

Projectively, write the selected endpoint lines in a common chart as

$$v_H(\omega, \tau), \quad v_\infty(\omega, \tau), \quad \delta = v_\infty - v_H.$$

At a simple root,

$$\frac{b(\omega_n)}{a'(\omega_n)} = \frac{\delta_\tau(\omega_n, 0)}{\delta_\omega(\omega_n, 0)}.$$

This ratio is invariant under analytic nonzero endpoint units and under moving the matching section.

10 Certified QNM specialization

The validated contour is oriented positively and encloses the disk

$$\Omega_n = \{\omega : |\omega - (-0.3736716844180418 + 0.0889623156889357i)| \leq 10^{-7}\}.$$

Panelwise projective enclosures exclude zero on the contour, include all chart-transition units, and give winding number +1. Thus the disk contains exactly one simple spin-two zero.

On the localized disk, the intrinsic tangent is enclosed by

$$\delta_\tau(\Omega_n) \subset [0.0010932 \pm 7.60 \times 10^{-8}] + i[-0.0002195 \pm 8.89 \times 10^{-8}],$$

which excludes zero. An independent local calculation gives a spin-one mismatch modulus greater than $1/2500$.

Theorem 10.1 (Certified defective Schwarzschild resonance). *The complete axial connection has Smith valuations*

$$(0, 0, 2)$$

at the unique QNM in Ω_n . The algebraic multiplicity of its spin-two divisor is two, its geometric multiplicity is one, and it has one length-two connection root chain.

Proof. The winding certificate gives one simple spin-two zero. The local tangent enclosure proves $b(\omega_n) \neq 0$, while the spin-one enclosure proves $g(\omega_n) \neq 0$. Proposition 9.1 applies. $\square$

11 Resonant evaluation of the self-extension

The projective selector, adjoint overlap, root-chain coefficient, and Green-pole residue are different representations of one resonant functional. This is the finite-interval Evans/Melnikov mechanism; compare the general relation between Evans derivatives and adjoint generalized eigenfunctions in [1].

Let

$$L(\omega) : X \longrightarrow Y$$

denote the analytic finite-interval spin-two boundary pencil, with the endpoint conditions included either in its domain or as finite-dimensional components of an augmented range. At a simple resonance choose right and left states

$$L_n u_n = 0, \quad L_n^* \tilde{u}_n = 0, \quad L_n = L(\omega_n),$$

and define, using the bilinear duality of the augmented boundary problem,

$$\alpha_n = \langle \tilde{u}_n, L'(\omega_n) u_n \rangle, \quad \alpha_n \neq 0.$$

Let the repeated block be

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix}, \quad \beta_n = \langle \tilde{u}_n, K(\omega_n) u_n \rangle.$$

All endpoint components of the augmented pencil and its duality are included in these pairings.

Theorem 11.1 (Resonant evaluation theorem). *In endpoint frames compatible with the partial jet,*

$$\kappa_n := \frac{b(\omega_n)}{a'(\omega_n)} = \frac{\beta_n}{\alpha_n} = -\omega'_n(0).$$

Here $\omega_n(\tau)$ is the resonance branch of

$$L(\omega, \tau) = L(\omega) + \tau K(\omega)$$

through ω_n . The equality is invariant under a common analytic nonzero Evans unit.

Proof. Differentiate

$$L(\omega_n(\tau), \tau)u(\tau) = 0$$

at $\tau = 0$:

$$L_n \dot{u} + \omega'_n(0)L'_n u_n + K_n u_n = 0.$$

Pairing with $\tilde{u}_n$ removes the first term and gives

$$\omega'_n(0)\alpha_n + \beta_n = 0.$$

On the connection side, differentiation of

$$a(\omega_n(\tau), \tau) = 0$$

gives

$$a'(\omega_n)\omega'_n(0) + \partial_\tau a(\omega_n, 0) = 0.$$

The exact partial-jet construction identifies $\partial_\tau a(\omega_n, 0) = b(\omega_n)$. Multiplication of the Evans coefficient by an analytic unit $u(\omega, \tau)$ multiplies both a' and b at the root by $u(\omega_n, 0)$, so the ratio is unchanged. $\square$

Proposition 11.2 (Resonant functional on the extension class). *The assignment*

$$\mathfrak{M}_n([K]) := \langle \tilde{u}_n, K(\omega_n)u_n \rangle$$

is a well-defined linear functional on the local self-extension class. Thus nonsplitting is the statement $[K] \neq 0$, while defectiveness at ω_n is the sharper statement $\mathfrak{M}_n([K]) \neq 0$.

Proof. Under a triangular change of splitting,

$$K \mapsto K + LQ - QL.$$

At the resonance,

$$\langle \tilde{u}_n, (L_n Q - Q L_n)u_n \rangle = \langle L_n^* \tilde{u}_n, Q u_n \rangle - \langle \tilde{u}_n, Q L_n u_n \rangle = 0.$$

When endpoint conditions are encoded in an augmented pencil, the same calculation includes the corresponding boundary terms by definition. $\square$

The theorem and proposition separate a global module property from its evaluation at one resonance. A nonzero extension class need not be defective at every QNM: the linear functional $\mathfrak{M}_n$ may vanish at isolated modes.

12 The generalized root vector is non-Einstein

Proposition 12.1 (Nonzero carrier in the generalized root). *Assume*

$$a(\omega_n) = 0, \quad a'(\omega_n) \neq 0, \quad b(\omega_n) \neq 0.$$

The geometric root vector belongs to the source Einstein submodule, while every length-two root polynomial has first generalized coefficient with nonzero projection to the carrier quotient. In the normalized triangular frame that projection is

$$-\frac{a'(\omega_n)}{b(\omega_n)} = -\frac{\alpha_n}{\beta_n} = -\frac{1}{\kappa_n}.$$

Thus the defective doubled resonance is not two copies of an Einstein quasinormal mode: its generalized member is intrinsically non-Einstein.

Proof. Put $z = \omega - \omega_n$ and write

$$a = a_1 z + O(z^2), \quad b = b_0 + O(z), \quad a_1 b_0 \neq 0.$$

For

$$c(z) = c_0 + z c_1 + O(z^2), \quad T_2(z) c(z) = O(z^2),$$

the order-zero equation gives $c_0 = e_X$, the source Einstein vector. Write $c_1 = x e_X + y e_Y$. The e_X component at order z is

$$a_1 + b_0 y = 0,$$

so $y = -a_1/b_0 \neq 0$. By Theorem 3.1, e_Y projects nontrivially to the Ricci-carrier quotient. Adding a multiple of e_X changes x , but cannot remove y . $\square$

13 Reduced outgoing Green double pole

Fix ordinary radii $2 < r_H < r_I < \infty$ inside the certified analytic continuation domains. Let

$$H(\omega) : \mathbb{C}^3 \longrightarrow C^\infty([r_H, r_I]; \mathbb{C}^6)$$

be the analytic Poisson operator for horizon-selected homogeneous data, let $\Gamma_-(\omega)$ be the unwanted incoming trace at r_I , and set

$$T(\omega) = \Gamma_-(\omega) H(\omega).$$

For a compactly supported source F , let $P_H(\omega)F$ denote the inhomogeneous solution with zero horizon data. The outgoing solution is

$$R_{\text{out}}(\omega)F = P_H(\omega)F - H(\omega)T(\omega)^{-1}\Gamma_-(\omega)P_H(\omega)F. \quad (3)$$

Theorem 13.1 (Reduced outgoing Green pole). *On a neighborhood of the certified frequency ω_n , the operator*

$$R_{\text{out}}(\omega) : C_c^\infty((r_H, r_I); \mathbb{C}^6) \longrightarrow C_{\text{loc}}^\infty((r_H, r_I); \mathbb{C}^6)$$

is meromorphic and has a nonzero rank-one pole of order two. Up to the endpoint trace sign convention, its principal coefficient is

$$\text{Coeff}_{(\omega - \omega_n)^{-2}} R_{\text{out}} = \frac{b(\omega_n)}{a'(\omega_n)^2} H(\omega_n) e_X \otimes [e_Y^* \Gamma_-(\omega_n) P_H(\omega_n)].$$

Proof. All factors in (3) except T^{-1} are analytic. After analytic elimination of the spin-one unit, the singular block is

$$T_2^{-1} = \begin{pmatrix} a^{-1} & -ba^{-2} \\ 0 & a^{-1} \end{pmatrix},$$

and therefore

$$\text{Coeff}_{z^{-2}} T_2^{-1} = -\frac{b(\omega_n)}{a'(\omega_n)^2} e_X \otimes e_Y^*.$$

The minus sign in (3) gives the displayed convention.

The range vector is nonzero because the homogeneous Poisson map is injective, so $H(\omega_n)e_X \neq 0$. The covector is nonzero by a cutoff Jost construction. Given any incoming coefficient q , choose the corresponding incoming Jost solution J_-q near r_I , multiply it by a smooth cutoff χ that vanishes near r_H and equals one near r_I , and set

$$F = (D - \mathbb{A}(\omega_n))(\chi J_-q).$$

Then F is compactly supported, $P_H(\omega_n)F = \chi J_-q$, and $\Gamma_- P_H(\omega_n)F = q$. Hence the incoming trace map from compact sources is onto. Taking $q = e_Y$ shows that the covector does not vanish. The tensor product is therefore nonzero and rank one. $\square$

Corollary 13.2 (Non-annihilation by certified reconstruction). *The certified rational reconstruction into the axial metric/curvature variables does not annihilate the range $H(\omega_n)e_X$. Hence at least one reconstructed local observable has the same order-two radial response pole.*

In the finite-interval Fredholm normalization of Section 11, the scalar resolvent has

$$L(\omega)^{-1} = \frac{u_n \otimes \tilde{u}_n}{\alpha_n(\omega - \omega_n)} + O(1).$$

The upper-right block of $\mathbb{L}^{-1}$ is $-L^{-1}KL^{-1}$, and Theorem 11.1 therefore gives the invariant principal coefficient

$$\boxed{\text{Coeff}_{(\omega - \omega_n)^{-2}}(\mathbb{L}^{-1})_{12} = -\frac{\beta_n}{\alpha_n^2} u_n \otimes \tilde{u}_n = -\frac{\kappa_n}{\alpha_n} u_n \otimes \tilde{u}_n.} \quad (4)$$

The Poisson/trace coefficient in Theorem 13.1 is the same rank-one residue expressed in endpoint coordinates.

14 Conditional global Fredholm promotion

Theorem 13.1 is a differential Green-operator theorem on a finite ordinary interval. A global physical resolvent requires a separate closed-operator realization. The correct abstract block notation is

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix},$$

where $L(\omega) : D \subset X \rightarrow Y$ is an analytic Fredholm pencil of index zero and $K(\omega) : D \rightarrow Y$ is graph-norm bounded. Suppose

$$u_n \in \ker L(\omega_n), \quad \tilde{u}_n \in \ker L(\omega_n)^*,$$

and define using the Y^* - Y bilinear duality

$$\alpha_n = \langle \tilde{u}_n, L'(\omega_n)u_n \rangle, \quad \beta_n = \langle \tilde{u}_n, K(\omega_n)u_n \rangle.$$

Proposition 14.1 (Conditional global pole formula). *If L^{-1} is meromorphic near ω_n , its zero is simple, $\alpha_n \neq 0$, and $\beta_n \neq 0$, then*

$$\mathbb{L}^{-1} = \begin{pmatrix} L^{-1} & -L^{-1}KL^{-1} \\ 0 & L^{-1} \end{pmatrix}$$

has an order-two pole with principal coefficient

$$-\frac{\beta_n}{\alpha_n^2} u_n \otimes \tilde{u}_n$$

in the upper-right block.

This proposition is standard block algebra once the Fredholm spaces and pairing exist. What remains open is not the finite-interval pole proved above, but the construction of:

1. closed weighted horizon and null-infinity domains;
2. a physical gauge quotient and bounded metric reconstruction;
3. a causal exterior resolvent on a Laplace contour;
4. a justified contour deformation and control of the non-pole part.

Only after these steps could the double pole be promoted to a rigorous $te^{i\omega_n t}$ contribution to retarded ringdown.

15 Critical Einstein–Weyl mass comparison

Theorem 6.1 reduces the local mass-comparison problem to one scalar. Starting from the covariant Einstein–Weyl parent system, one must:

1. derive the finite-mass axial first-order system;
2. eliminate its constraints in a gauge compatible with the massless Regge–Wheeler companion frame;
3. differentiate at $m^2 = 0$;
4. compute its projective cocycle $\mathcal{I}_{\text{mass}}$;
5. determine the coefficient $c(\omega)$ in

$$[\mathcal{I}_{\text{mass}}] = c(\omega)[f].$$

Nonzero $c(\omega)$ proves local equivalence of the repeated blocks. The stronger scattering identity

$$b = -\partial_{m^2} a|_{m^2=0}$$

additionally requires equality of the complete filtered deformation, moving massive horizon and infinity phases, and endpoint normalization. The QNM-slope identity requires, in addition, the global Fredholm realization of Section 14. None of these stronger identifications is inferred solely from the normal form.

The resonant functional makes the remaining local calculation especially small. From Theorem 6.1,

$$\beta_n^{\text{Bach}} = \frac{i\omega_n}{2} \mathfrak{M}_n([f]).$$

The certified inequality $\kappa_n \neq 0$ implies

$$\mathfrak{M}_n([f]) \neq 0.$$

If a finite-mass deformation has

$$[\mathcal{I}_{\text{mass}}] = c_m(\omega)[f], \quad c_m(\omega_n) \neq 0,$$

then its local resonance velocity is necessarily nonzero and is related to the Bach selector by

$$\left. \frac{d\omega_n}{dm} \right|_{m=0} = -\frac{2c_m(\omega_n)}{i\omega_n} \kappa_n,$$

with m denoting the declared deformation parameter. Endpoint compatibility is still required before this local formula is identified with a physical massive Jost branch.

Exact parent/radial comparison target

Let R reconstruct the tensor carrier from the scalar master field and let P project the tensor equation back to the scalar equation. The desired comparison is the augmented-pencil identity

$$K_{\text{radial}} - PAR = LQ - QL,$$

possibly with an explicitly recorded endpoint coboundary. If this identity is proved, Proposition 11.2 gives immediately

$$\beta_n^{\text{radial}} = \langle \tilde{u}_n, PARu_n \rangle = \beta_n^{\text{parent}}.$$

Accordingly, a separate numerical equality test is not the primary target: the correct promotion gate is an exact cohomological projection theorem.

16 Computational certificates and claim boundary

The computer-assisted inputs are separated from the exact derivations above.

Certificate family	Claim used here	Evidence type
Axial triangular preflight	Exact RW/RW/Maxwell filtration and reconstruction	Exact rational
Projective cocycle	Gauge law and nontrivial extension class	Exact rational
RW/Maxwell simplicity	Physical-axis simplicity and scalar endomorphisms	Exact rational
Full Evans contour	Zero-free boundary and winding +1	Validated interval
Local selector	$\delta_\tau \neq 0$ on the root disk	Validated interval
Spin-one local unit	Full three-factor Smith type $(0, 0, 2)$	Validated interval
Fredholm promotion	Finite-interval rank-one radial Green pole	Exact reduction plus validated inputs

The theorem statements and proofs are the mathematical authority. The machine-readable records pin exact witnesses, interval margins, source hashes, and independently executable checks. A missing input, unsupported domain, timeout, or failed mutation causes rejection rather than partial acceptance.

Explicit non-results. This paper does not establish a causal spacetime resolvent, a complete QNM expansion, a late-time asymptotic formula, a $te^{i\omega_n t}$ term, a time-domain instability, a physical Einstein–Weyl mass identification, an all- ℓ Bach nonsplitting theorem, or a quantum positivity statement.

17 Conclusion

The axial Schwarzschild Bach equation contains a repeated Regge–Wheeler factor, but the repetition is not a rational direct sum. It is a non-split self-extension whose projective class is represented by the elementary direction $(i\omega/2)(1 - 2/r)$. At the certified damped mode, the scalar zero is simple while the extension selector is nonzero. The resulting defect has concrete differential content: the generalized root has a nonzero Ricci-carrier quotient, and the reduced outgoing Green operator has a genuine rank-one double pole.

This separates three notions that should not be conflated. A repeated scalar divisor fixes algebraic multiplicity; the extension selector decides whether that multiplicity is semisimple or defective; and a causal time-domain interpretation requires a further global Fredholm theory. The first two questions are settled here at one physical Schwarzschild resonance. The third remains a sharply delimited next problem.

References

- [1] T. Kapitula, The Evans function and generalized Melnikov integrals, *SIAM Journal on Mathematical Analysis* **30** (1999), 273–297, [doi:10.1137/S0036141097327963](https://doi.org/10.1137/S0036141097327963).